The cell: the unit of life

What is a cell?

The cell is the smallest living unit of a living thing. Every living thing is made of one cell or of many cells. The human body has a huge number of cells that work together.

Most cells are very small. We need a microscope to see them.

The main parts of a cell

Most cells have three main parts: the cell membrane, the cytoplasm and the nucleus.

The cell membrane

The cell membrane is a thin cover around the cell. It lets in what the cell needs, such as water and food. It lets waste out. So the membrane controls what enters the cell and what leaves it.

The cytoplasm

The cytoplasm is a jelly-like material that fills the cell. Small parts called organelles float in it.

The nucleus

The nucleus is the control centre of the cell. It keeps the genetic material, which carries the instructions for the traits of the living thing. The nucleus directs the work of the cell, such as growth and division.

The mitochondria

Mitochondria are small organelles in the cytoplasm. Their job is to produce the energy the cell needs. They take in food and oxygen, and they turn the energy stored in food into energy the cell can use. This process is called cellular respiration. That is why mitochondria are called the powerhouse of the cell.

Cells that do a lot of work need more energy. So muscle cells have many mitochondria.

Plant cells and animal cells

Both have a membrane, cytoplasm, a nucleus and mitochondria. A plant cell has some extra parts:

- The cell wall: a stiff wall outside the membrane. It supports the cell and gives it its shape.

- Chloroplasts: they make food from sunlight, water and carbon dioxide. This is photosynthesis.

- A large vacuole: it stores water.

Summary

The cell is the unit of life. The membrane controls what goes in and out. The nucleus keeps the genetic material. The mitochondria produce energy. A plant cell also has a cell wall, chloroplasts and a large vacuole.